

Graduate Algorithms

Quiz 1

August 10, 2026

Curiosity is love for the world, and when you love something, you want to get to know it more. So when you share facts about crows or pterodactyls or whatever, what you are expressing is gratitude for the ability to exist in and come to understand the beauty of the world around you.

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Instructions

- Total marks: 30. Time: 45 minutes.
- There are 4 problems; attempt any 3 problems.
- Appendix A at the end contains reference material you may use.

Question 1: Gold, Stone and Strategy

10 marks

King Midas and Medusa alternatively play on an 54×54 chessboard. Initially, all of the cells of the chessboard are made of wood. In each of his turn, Midas chooses three squares which are in the same column or row, and turns them into gold. In each of Medusa's turn, she can choose any 2 rows and 2 columns and turns the 4 intersecting cells into stone. King Midas wins if he can make a 2 by 2 grid square entirely of gold. Medusa wins if she can always prevent Midas doing so. Who wins? (Note: Midas can't turn stone to gold and not can Medusa turn gold to stone)

Question 2: The Token Pile

10 marks

Ada and Babbage play the following game. A pile of 3,141,592 tokens sits on the table. They move alternately, Ada first. On a turn, a player must do exactly one of the following:

- (i) remove a single token from the pile; or
- (ii) remove exactly half the tokens, rounding *down* when the pile size is odd (so a pile of n tokens loses $\lfloor n/2 \rfloor$ of them).

The player who leaves the pile with exactly one token wins. Assuming both players play as well as possible, which of them has a winning strategy? Prove that your answer is correct.

Question 3: Recurrences

7+3 marks

Consider the recurrence

$$T(n) = 5T\left(\frac{n}{3}\right) + \Theta(n), \quad T(n) = \Theta(1) \quad \text{for } n \leq 3.$$

- (a) Using the substitution method, prove that $T(n) = O(n^{\log_3 5})$. State your inductive hypothesis explicitly, carry out the inductive step, and say exactly which constants you need and how large they must be. *Hint:* a guess of the bare form $c n^{\log_3 5}$ will not close the induction. See Appendix A.1.

- (b) Verify your answer using the Master theorem (Appendix A.2). Identify a , b and $f(n)$, say which case applies and why, and state the resulting bound.

Question 4: Counting Triangles

10 marks

We have an algorithm (refer to **algorithm 1**) to count the number of triangles in a graph. Could you calculate the running time for the given algorithm, assuming checking if an edge exists is $O(1)$?

Algorithm 1 An algorithm to count the number of triangles in a graph

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1: count  $\leftarrow$  0
2: for each vertex  $v \in V$  do
3:   for each pair  $u, w \in N(v)$  such that  $u \neq w$  do
4:     if  $uw \in E$  then
5:       count  $\leftarrow$  count + 1
6:     end if
7:   end for
8: end for
9: return count/3

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Appendix A: Reference material

A.1 The substitution method

The method has two steps: guess the form of the solution, then verify it by induction on n , solving for the constants along the way. Three things to keep in mind.

- You must establish the *exact* inequality you guessed, with one fixed constant, e.g. $T(n) \leq cn^\alpha$ for all $n \geq n_0$. Ending the inductive step with an expression that is merely $O(n^\alpha)$ does not count: the constant must not grow with n .
- If the induction fails by a lower-order term, strengthen the hypothesis by subtracting one, e.g. guess $T(n) \leq cn^\alpha - dn$. A stronger hypothesis gives you a stronger statement to use inside the inductive step, which is what absorbs the additive cost.
- Base case: asymptotic bounds need only hold for $n \geq n_0$, so choose n_0 large enough that the recurrence applies and the inductive step is valid, and choose the constants large enough to cover the finitely many values below n_0 .

A.2 Master theorem

Let $a \geq 1$ and $b > 1$ be constants, let $f(n)$ be a nonnegative function, and let

$$T(n) = aT\left(\frac{n}{b}\right) + f(n).$$

Compare $f(n)$ with $n^{\log_b a}$:

1. If $f(n) = O(n^{\log_b a - \varepsilon})$ for some constant $\varepsilon > 0$, then $T(n) = \Theta(n^{\log_b a})$.
2. If $f(n) = \Theta(n^{\log_b a})$, then $T(n) = \Theta(n^{\log_b a} \log n)$.
3. If $f(n) = \Omega(n^{\log_b a + \varepsilon})$ for some constant $\varepsilon > 0$, and if $a f(n/b) \leq k f(n)$ for some constant $k < 1$ and all sufficiently large n , then $T(n) = \Theta(f(n))$.